

**Q1**

An algorithm to get compute a  $x^n$  is below.

```
def x_to_the_power_of_n( x, n ):

    m = n
    y = 1
    z = x

    if ( m%2 ):
        y = y * z
        m = m//2

    while( m >= 1):
        z = z * z
        if ( m%2 ):
            y = y * z
            m = m//2

    return y
```

We measure complexity of this algorithm by the number of  $*$  operations.

1. For a given  $n$ , what is the minimum complexity of the algorithm?
2. For a given  $n$ , what is the maximum complexity of the algorithm?
3. For a random  $n$ , what is the expected complexity of the algorithm? (Define what “random” means.)

**Q2**

Given two integers  $a$  and  $b$ , prove that the length of the repetend of  $\frac{a}{b}$  is finite. (The reasoning works in any base but let us work with base 10 for this problem.)

**Q3**

We consider a floating-point system with 48 bits and we work with 10-bit exponent and 39-bit mantissa. We use a similar convention as IEEE-754 for 32-bit float or 64-bit double.

1. Give  $x_{huge}$ , the largest finite machine number.
2. Give  $x_{tiny}$ , the smallest positive machine number.
3. Give  $\varepsilon$ , that is the number such that  $1 + \varepsilon$  is the smallest machine number larger than 1.
4. Give  $u$ , the unit roundoff, that is the supremum for  $|fl(x) - x|/|x|$  for any real number  $x$  in  $[-x_{huge}, -x_{tiny}] \cup [x_{tiny}, x_{huge}]$ .

**Q4**

(Problem 4.6 in Higham) (Shewchuk [1038, 1997]) Consider correctly rounded binary arithmetic. Show that if  $a$  and  $b$  are floating point numbers then  $\mathbf{err}(a, b) = a + b - fl(a + b)$  satisfies  $|\mathbf{err}(a, b)| \leq \min(|a|, |b|)$ . Hence show that, barring overflow in  $fl(a + b)$ ,  $\mathbf{err}(a, b)$  is a floating point number.

**Q5**

(Problem 2.27 in Higham) (Kahan) Show that with the use of a fused multiply-add operation the algorithm

$$\begin{aligned}w &= be \\e &= w - be \\x &= (ad - w) + e\end{aligned}\tag{1}$$

computes  $x = \det\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)$  with high relative accuracy.